

Chapter 151 — Minimally Invasive and Endoscopic Management of Benign Prostatic Hyperplasia

- **Urinary obstruction in men is ancient:** the Chinese catheterised with onion stalks; the Greeks and Romans used wooden or metal tubes. The cause of obstruction was not understood.
- **Herophilus (c. 300 BCE)** first used the word *prostate* (reported by Galen). **Jean Riolan (1649)** first suggested that prostatic enlargement causes urinary retention.
- **Morgagni (1760)** gave one of the earliest descriptions of benign prostatic hyperplasia (BPH): an autopsy of a 73-year-old man with a thick-walled bladder and an obstructing pear-shaped prostate.
- **18th-century surgery was blind:** punch-type tools to incise and remove tissue; sounds to dilate the urethra (some with concealed blades to incise the bladder neck or lobes); lance-shaped devices to puncture obstructing tissue.
- **Bozzini's cystoscope (1806)** first allowed visualisation of the lower urinary tract.
- **Early 1900s operations:** perineal simple prostatectomy (Hugh Hampton Young, 1903); open suprapubic prostatectomy (John Thomas Walker, 1919); transurethral incision with Bottini's electric knife. All were **morbid and unreliable**, which drove the search for a better technique.
- **New energy sources and devices** (morcellators, prostatic urethral lift, temporary implanted prostatic devices) now give more surgical options for lower urinary tract symptoms (LUTS) attributed to BPH than ever. These include the **minimally invasive surgical therapies (MISTs)**.
- **Novel therapies must be judged by consistent, clearly defined criteria.** Placebo effects are considerable where design lacks rigour (retrospective, single-cohort, uncontrolled). Premature enthusiasm for **transurethral needle ablation of the prostate (TUNA)** produced American Urological Association (AUA) guideline recommendations that were **later retracted**.

Surgical Trends

- **Transurethral resection of the prostate (TURP) peaked in 1987** (Medicare data). The fall since then is attributed to better medical therapy, the rise of MISTs and falling reimbursement. By 1997 the TURP rate was less than half its peak.
- **TURP still dominated (about four-fifths of BPH procedures) in 1999**, while the total number of BPH operations kept falling.
- **Total BPH surgery rose again from 2002**, driven entirely by a steep rise in MISTs: transurethral microwave therapy (TUMT), TUNA and transurethral laser coagulation or vaporisation. TURP kept declining year on year.
- **Total BPH surgery peaked in 2005**, then declined again. TURP continued to fall while laser vaporisation rose to nearly match it by 2008. Total surgical BPH cases fell further between 2008 and 2014.
- **Operations in the 2023 AUA Guidelines:**
 - Transurethral incision of the prostate (TUIP), TURP, and transurethral vaporisation of the prostate (TUVF).
 - Photoselective vaporisation of the prostate (PVP) and prostatic urethral lift (PUL; UroLift).
 - Water vapour thermal therapy (WVTT; Rezūm), temporary implanted prostatic devices (TIPD; iTind) and robotic waterjet treatment (RWT; Aquablation).
 - Holmium laser enucleation of the prostate (HoLEP) or thulium laser enucleation of the prostate (ThuLEP).
 - Simple prostatectomy (open, robotic or laparoscopic) and prostate artery embolisation (PAE).

Peer-reviewed comparisons of current trends across all of these are limited.

- **American Board of Urology case logs (2008–2021):**
 - **TURP remained the most performed BPH procedure** and rose year on year.
 - **HoLEP use was flat.** It was more likely among high-volume surgeons and endourologists.
 - **PUL rose steeply after its 2015 introduction**, from a small fraction of cases to about one-third by 2020.

Concept — *Surgical volume for BPH follows medical therapy, reimbursement and the arrival of new technology rather than disease burden, which is why TURP falls while MISTs and lasers rise.*

PROSTATE SIZE/VOLUME

- **Prostate size increases with age.** Community cohorts show median volume rising from the low 20s cc in the fifth decade to the low-to-mid 30s cc by the eighth. The **Olmsted County study** estimated an average annual growth of about 1.6%; the Flint Men's Health Study showed the same age association.
- **Growth is faster in men with symptomatic BPH.**
 - In the **Medical Therapy of Prostatic Symptoms (MTOPS)** study, prostate size rose by a median of 24% over 4.5 years (about 5.3% per year).
 - The **Proscar Long-term Efficacy and Safety Study (PLESS)** showed a strong correlation between serum prostate-specific antigen (PSA) and prostate size, and 14.1% growth over 4 years (about 3.5% per year).
- **Size alone does not determine symptom severity.** Correlation between size and the International Prostate Symptom Score (IPSS) is weak or absent.
- **Size does predict the need for surgery.**
 - **Larger glands carry a higher risk of acute urinary retention (AUR).** In Olmsted County, a prostate >30 cc tripled the risk of AUR.
 - In the placebo arm of the **Reduction by Dutasteride of Prostate Cancer Events (REDUCE)** trial, a gland of 40.1–80 cc carried a 67% higher risk of developing LUTS than a gland ≤40 cc.

KEY POINTS

- Medical management and new technologies have changed which surgical treatments are used for LUTS secondary to BPH.
- Medical therapy may defer surgery, so surgical patients are more complex.
- Larger prostate size is associated with the need for BPH surgery.

Concept — *Volume predicts **progression** (retention, surgery) but not **symptoms**; size selects the operation, while symptoms decide whether to operate.*

WORKUP

- **Initial evaluation (AUA Statement 1):** history, physical examination, IPSS and urinalysis.
- **IPSS (also called the AUA Symptom Score, AUASS)** quantifies baseline LUTS and documents response to treatment. Severity bands: **0–7 mild, 8–19 moderate, 20–35 severe.**
- **Assess comorbidities and medications that affect voiding**, to guide behavioural modification and first-line medical therapy. Medical management is covered in Chapter 150.
- **Indications for surgery (Statement 26):**
 - **Renal insufficiency** secondary to BPH.
 - **Refractory urinary retention** secondary to BPH.
 - **Recurrent urinary tract infections (UTIs).**

- **Recurrent bladder stones or gross haematuria** secondary to BPH.
- **LUTS refractory to other therapy**, or a patient unwilling to use it.
- **Presurgical testing:**
 - **Should** include a post-void residual (PVR) (Statement 6).
 - **Consider** assessment of prostate size and shape by transrectal or abdominal ultrasound, cystoscopy, or cross-sectional imaging such as magnetic resonance imaging (MRI) or computed tomography (CT) (Statement 5).
 - **Consider** uroflowmetry (Statement 7).
 - **Consider pressure-flow studies** when there is diagnostic uncertainty (Statement 8).
- **Counselling:** warn of possible treatment failure and the need for additional or secondary treatment before any surgery or MIST (Statement 9).
- **Units of size:** cc, mL and g are used interchangeably because prostate tissue has a specific gravity of about 1.05 g/mL. **Cubic centimetres are technically correct**, because volume is calculated with the ellipsoid formula $\pi/6 \times (L \times W \times H)$.
- **Prostate cancer screening belongs in the BPH workup**, although the BPH guideline does not address it directly.
 - The AUA 2023 Early Detection Guideline recommends **shared decision-making, with PSA screening at ages 50–69 every 2–4 years** (Statements 1, 2 and 6). The interval is then personalised, or screening stopped, according to age, comorbidity, PSA, risk, preference and life expectancy (Statement 7).
 - **An elevated PSA warrants biopsy before BPH surgery.**
 - The European Association of Urology (EAU) recommends **prebiopsy MRI**: a normal MRI avoids about 30% of biopsies at the cost of missing some grade group ≥ 2 cancers.
 - Detecting cancer first matters because **clinically significant cancer needs radical prostatectomy, not TURP.**
- **Summary of counselling:** weigh comorbidities, the contribution of BPH to the overall voiding pattern, the need for treatment, the probability of success and the risks. **The patient's goals guide the choice.**

DEFINING OUTCOMES

Subjective

- **Administer the IPSS at baseline (Statement 1) and repeat it after treatment (Statement 3).**
- **Other validated instruments:**
 - Michigan Incontinence Symptom Index (MISI) for incontinence.
 - Sexual Health Inventory for Men (SHIM) for erectile function.
 - Male Sexual Health Questionnaire Ejaculatory Dysfunction short form (MSHQ-EjD) for ejaculatory and orgasmic function.

Before-and-after use makes subjective outcomes reproducible.

- **The sham effect is large.** A systematic review found that sham endoscopic surgery produced a meaningful improvement in IPSS and in maximum flow rate (Qmax) at 3 months. This is why **sham-controlled trials are essential** for new devices.

Objective

- **Standard objective end points:** AUR occurrence, Qmax, PVR, weight of tissue removed (where available), and change in prostate volume and/or PSA.

- Objective data depend on accurate reporting and are **confounded by loss to follow-up** and by non-responders receiving additional treatment.

Need for Secondary Treatment

- **Treatment failure is an important end point:** it guides patient choice and drives cost.
- **It is not clear-cut.** The threshold and type of retreatment vary by surgeon, patient, category of failure (objective, subjective or both) and initial modality. They also depend on how completely the obstructing adenoma was treated.
- **Reported retreatment rates depend on duration and completeness of follow-up.** Older trials often omitted retreatment with medication; newer trials report it.

Antibiotic Coverage

- **AUA Best Practice Statement (BPS) on antimicrobial prophylaxis, updated 2019:**
 - Consider periprocedural systemic prophylaxis for any invasive procedure, in light of patient risk and allergy (Statement 3).
 - **A single dose is appropriate** for most uncomplicated cases (Statement 5) and for clean-contaminated genitourinary procedures (Statement 7).
- **Transurethral procedures should cover gram-negative enteric organisms and enterococci.** Examples are first- or second-generation cephalosporins or trimethoprim-sulfamethoxazole.
- **Tailor the choice** to the local antibiogram (Statement 12) and to any preoperative urine culture (Statement 16).
- **Active symptomatic UTI: defer transurethral surgery** until treatment is complete and symptoms have resolved (Statement 17).

Histologic Specimen

- **Before PSA screening, TURP chips frequently diagnosed prostate cancer:** about a quarter of cases were incidental cancers. This fell sharply after PSA screening began.
- With a **normal PSA and digital rectal examination, incidental cancer in BPH specimens is uncommon.** The diagnosis of pT1 and pT1b tumours fell markedly during the PSA era.
- Surveillance, Epidemiology, and End Results (SEER) data: with a **PSA cut-off of 4 ng/mL, very few clinically significant cancers (Gleason ≥ 7) would be missed.**
- **Therefore, few significant cancers are missed** by procedures that yield no specimen: TUVP, PVP, PUL, WVTT, PAE and RWT.

Matching the Treatment With the Patient

- **Factors to weigh:**
 - The risks, benefits and safety profile of each treatment, and the surgeon's own experience.
 - **Prostate size and shape.**
 - A history of **retention.**
 - **Inability to stop anticoagulation.**
 - Patient preference, for example **preserving antegrade ejaculation.**
- **The 2021/2023 AUA Guidelines introduced size-based recommendations** (Fig. 151.1). Earlier versions had none.
- **Cystoscopy is best for identifying a median lobe** or excluding a stricture or bladder tumour. It is **unreliable for sizing.** A rule of thumb is that each centimetre of prostatic urethra beyond the normal 2.5 cm represents about 10 g of extra tissue, but ultrasound, CT and MRI are more accurate.

- **Higher bleeding risk (unable to hold anticoagulation) → HoLEP, ThuLEP or PVP (Statement 43).**

Otherwise, hold antiplatelet or anticoagulant drugs according to standard practice after conferring with the prescriber.

FIG. 151.1 SURGICAL MANAGEMENT ALGORITHM FOR LUTS SECONDARY TO BPH (AUA BPH 2023 GUIDELINES), REBUILT

Prostate volume	Guideline options	Restrictions within the band
Small, <30 cc	HoLEP, PVP, ThuLEP, TIPD, TUIP , TURP, TUVP	TUIP only ≤30 cc ; TIPD 25–75 cc, no obstructing middle lobe
Average, 30–80 cc	RWT, HoLEP, PVP, PUL , ThuLEP, TIPD, TURP, TUVP, WVTT	RWT 30–80 cc; PUL 30–80 cc and no obstructing midline tissue ; WVTT 30–80 cc; TIPD 25–75 cc, no obstructing middle lobe
Large (>80–150 cc) or very large (>150 cc)	Simple prostatectomy (open, laparoscopic or robotic), HoLEP , ThuLEP	—
Medically complicated / higher bleeding risk	HoLEP, PVP, ThuLEP (lower transfusion need)	See the International Consultation on Urological Diseases (ICUD)/AUA review on anticoagulant and antiplatelet therapy
Wishes to preserve erectile and ejaculatory function	PUL or WVTT	Within their 30–80 cc criteria

- **What the algorithm tests:**
 - Only enucleation (HoLEP, ThuLEP) spans every size band.
 - Only simple prostatectomy joins it above 80 cc.
 - The MISTs and RWT sit in the 30–80 cc band. PUL and TIPD also require no obstructing median lobe.
- **Trap: TUIP is the only option confined to small glands**, and PVP, TURP and TUVP stop at 80 cc.

KEY POINTS

- At initial evaluation: history, physical examination, urinalysis and IPSS.
- Consider surgery for renal insufficiency, refractory retention, recurrent UTIs, recurrent stones or gross haematuria secondary to BPH, or LUTS refractory to (or patient unwilling to use) medication.
- Give perioperative antimicrobial prophylaxis guided by local antibiograms and any preoperative urine culture.
- Counselling considers surgeon and patient factors: preferences, prostate shape and size, and comorbidities.

Concept — *The 2023 guideline turns operation choice into a size-and-shape decision modified by bleeding risk and sexual-function priorities. That is why imaging or cystoscopic assessment of volume and a median lobe precedes every recommendation.*

SPECIFIC TREATMENTS — MINIMALLY INVASIVE SURGICAL THERAPIES

Minimally Invasive Surgical Therapies

- **The ideal MIST:**
 - Office-based, in a single session, **without general or regional anaesthesia or a hospital stay**.
 - Gives **better short-term benefit than medication**.
 - Avoids the sexual side effects and prolonged recovery of definitive surgery.
 - Removes the need for, and cost of, lifelong daily BPH medication.

Transurethral Microwave Therapy

- **TUMT was removed from the 2023 AUA Guidelines** as a legacy technology.

Prostatic Urethral Lift

Introduction

- **PUL (UroLift) was developed in 2004 and approved by the U.S. Food and Drug Administration (FDA) in 2013.** It alters prostatic anatomy **without tissue ablation**. Its advantages are low complication rates and minimal effect on erectile and ejaculatory function.
- **Permanent transprostatic implants** are delivered cystoscopically. They compress the parenchyma to open the prostatic urethra.
- **AUA: PUL for 30–80 cc without a median lobe.** Confirm the absence of a median lobe by cystoscopy or ultrasound, and verify the volume.
- **The FDA extended approval to ≤100 cc in 2020.** This is not yet in the AUA Guidelines because peer-reviewed prospective data are lacking.

Mechanism of Action

- **Performed under local anaesthesia** with cystoscopic guidance.
- **Implant design:** spring-loaded, with T-shaped ends.
 - **Urethral end: stainless steel**, which resists encrustation.
 - **Capsular tab: nitinol.**
 - **Connecting suture: permanent monofilament polyethylene terephthalate**, held under tension.
- **Deployment:** a 19-gauge hollow needle deploys each implant.
 - Glandular tissue is more compliant than the capsule, so implant tension opens the urethra.
 - Typically **4–6 implants are placed anterolaterally from bladder neck to apex** (Fig. 151.3). They sit in the anterolateral position, between the dorsal veins and the neurovascular bundles.
- **Beyond mechanical compression**, canine studies show localised atrophy, fibrosis and scarring within 12 months.
- **Implants are MRI-conditional.** Conditions: ≤3.0 T; maximum spatial field gradient 1500 G/cm; whole-body average specific absorption rate 4 W/kg for 15 min. The expected maximum implant heating is 2.4 °C.

Outcomes

THE TWO PUL RANDOMISED TRIALS — LIFT AND BPH6

	LIFT (2013)	BPH6 (2015)
Design	Multicentre randomised controlled trial (RCT), PUL vs sham , 2:1, 19 centres (US, Canada, Australia)	Multicentre RCT, PUL vs TURP , 1:1, 10 European centres; first RCT of PUL vs TURP
Population	Age ≥50, IPSS ≥13, Qmax ≤12 mL/s, 30–80 cc	Age ≥50, IPSS >12, Qmax ≤15 mL/s, PVR <350 mL, ≤60 cc
Primary end point	IPSS improvement vs sham at 3 months	Composite: LUTS relief, recovery quality, erectile, ejaculatory and continence preservation, safety; powered for non-inferiority (Δ 10%)
Key result	IPSS roughly halved; improvement about double that of sham; Qmax +4 mL/s at 12 months; about two-thirds needed no catheter; nearly all done under local anaesthesia	Composite met by 52.3% PUL vs 20.0% TURP: non-inferior and superior; 100% vs 60.6% antegrade ejaculation; faster recovery with PUL
Where TURP wins	—	Greater IPSS, Qmax and PVR improvement with TURP (IPSS and Qmax

LIFT (2013)		BPH6 (2015)
		persist at 2 years; PVR difference lost). Quality of life (QoL) equal
Durability / retreatment	5% at 1 year; 13.6% surgical retreatment at 5 years ; about one-third if medication restart and encrustation procedures are counted; improvements durable at 5 years; most patients satisfied	At 2 years: 13.6% PUL vs 5.7% TURP
Sexual function	No new sexual or ejaculatory dysfunction; none over 5 years	Erectile function maintained in both arms; MSHQ-EjD function better with PUL, but both scores equal
Limitations	Loss to follow-up by 5 years (per-protocol and intention-to-treat analyses agreed)	Powered only for the primary end point; composite not validated; self-selection of men indifferent to ejaculation

- **What the pair tests:** PUL beats sham and beats TURP on a **composite weighted towards recovery and sexual function**. TURP wins every pure efficacy measure (IPSS, Qmax, PVR) and has lower retreatment.
- **Trap:** "superior to TURP" in BPH6 refers to the composite end point only.

Prostatic Urethral Lift for Large Prostates and Those With a Median Lobe

- **The FDA approval for ≤100 cc (2020)** rested on a retrospective chart review that has not been made public. The **AUA found insufficient evidence for 81–100 cc**.
- **MedLift:** a prospective cohort of men with an obstructing median lobe treated with PUL showed improved IPSS, QoL and Qmax. It was **excluded by the AUA** because it was non-randomised with historical LIFT controls.
- A later retrospective median-lobe series had the same weakness, using historical BPH6/LIFT controls.

Complications

- **Short convalescence:** return to preoperative activity in about 8–9 days in LIFT. Because the procedure is non-ablative and leaves the mucosa intact, dysuria is minimal.
- **Mild, short-lived early symptoms:** dysuria, haematuria, pain and urgency, resolving within 2 weeks. Isolated events included clot retention on restarting warfarin and one bladder stone unrelated to the implant.
- **Encrustation:** a minority had exposed, encrusted implants in the bladder at follow-up cystoscopy.
 - **Deploy implants >1.5 cm distal to the bladder neck.**
 - Implants placed too proximally can be removed with endoscopic graspers.
- **Network meta-analysis:** a lower risk of major adverse events (AEs) than TURP.
- **Rare severe events** (Manufacturer and User Facility Device Experience, MAUDE, database):
 - Transfusion and clot evacuation.
 - **Pelvic or retroperitoneal haematoma;** arterial embolisation, orchidectomy and laparotomy in single cases.
 - Device malfunction: misfire, needle fragmentation, failed retraction and misplaced implants.
- **No reports of new erectile or ejaculatory dysfunction.** The bladder neck is preserved, so antegrade ejaculation is maintained.
- **Retreatment risk is higher than after TURP** (meta-analysis).

Conclusion

- **PUL: 30–80 cc, no median lobe.** It is durable to 5 years and preserves erectile and ejaculatory function. It suits men unwilling to take daily medication.

- **Retreatment (surgical or medical) is higher than after TURP**, so warn younger men that further treatment may be needed as BPH progresses.

Concept — *PUL moves tissue rather than removing it. That single fact explains its preserved bladder neck and ejaculation, its minimal dysuria, its median-lobe exclusion, and its higher retreatment as the gland keeps growing.*

CONVECTIVE RADIOFREQUENCY WATER VAPOUR THERMAL THERAPY

Introduction

- **WVTT (Rezūm) uses convective energy.** Heat is transferred through fluid. TUMT and TUNA used conductive energy, which is direct transfer of heat.
- **Convection delivers lower energy for a shorter time** to cause necrosis.
- **Mechanism:** radiofrequency (RF) power generates water vapour, which is injected transurethrally into the transition zone.
- **Prostatic collagen contains the thermal energy within the transition zone**, preventing off-target ablation.
- **AUA: WVTT for 30–80 cc**, and for men who want to preserve erectile and ejaculatory function (Statements 36 and 37).
- **Unlike PUL, WVTT can treat a median lobe.**

Mechanism of Action

- **Equipment:** a generator and a handheld delivery device (Fig. 151.4). The device includes a rigid cystoscope and an 18-gauge retractable needle with 12 vapour-emitting holes. Saline irrigation provides vision and cools the urethra.
- **Sequence:** needle penetrates a fixed 10.25 mm → first injection 1 cm distal to the bladder neck at 3 and 9 o'clock → further injections about 1 cm apart, distally to the proximal edge of the verumontanum → contiguous overlapping lesions (Fig. 151.5).
- **Each injection: vapour at 103 °C for 9 seconds**, at slightly above interstitial pressure, so that it disperses through the interstitium by convection.
- **Condensation releases 540 cal/g** onto cell membranes. This causes necrosis and a 1.5–2.0 cm lesion, confirmed on MRI and pathology.
- **Energy per procedure:** WVTT uses only 4 kJ. Comparators: TUNA 20 kJ, TUMT 86 kJ, ablative lasers 250 kJ.
- **Median lobe:** the procedure can be modified to treat it.
- **Outpatient procedure** under oral sedation, conscious sedation or prostatic block. Mean treatment time is about 5 minutes.

Outcomes

Comparative Studies — Rezūm II

- **Rezūm II:** a multicentre RCT, 2:1 WVTT vs sham (rigid cystoscopy with simulated treatment sounds).
 - Inclusion: age ≥50, IPSS ≥13, Qmax 5–15 mL/s, 30–80 cc, PVR ≤250 mL. **Median lobes were included.**
 - Primary end point: IPSS reduction at 3 months.
- **At 3 months:**
 - IPSS fell by 11.2 points with WVTT vs 4.3 with sham.
 - Qmax rose by 6.2 vs 0.5 mL/s, and QoL improved more.
 - **No PVR benefit.**
 - After unblinding, most sham patients crossed over.

- **Procedure details:** injections averaged about 4–5 into the lateral lobes and 1–2 into the median lobe.
 - Pain scores were higher than with sham.
 - Most patients had oral sedation only; a minority had a prostatic block or intravenous sedation.
 - Nearly all were catheterised for a few days, but **most catheters were discretionary**; only about a third followed a failed void trial.
 - Median return to usual activity was 4 days after catheter removal.
- **Durability:**
 - Around 50% improvement in IPSS and QoL, and Qmax, at 1 and 2 years, including the crossover group.
 - At 2 years there was no significant loss of erectile function, and ejaculatory bother improved.
- **Five years:**
 - **IPSS –48%, QoL –45%, Qmax +49%.**
 - The responder end point (no retreatment and $\geq 30\%$ IPSS fall) was met by the majority.
 - **No de novo erectile dysfunction (ED).**
 - **Surgical retreatment 4.4% and medical retreatment 11.1%**, most within the first 2 years.
- **Trap — the retreatment denominator:** 4.4% was calculated on the original 135 patients, not on the 77 followed to 5 years, which would give 7.8%. With 58 patients lost to follow-up, **true retreatment may be higher.**

Complications

- **AEs were mild to moderate.** Only two were serious: de novo prolonged retention, and nausea or vomiting after oral sedation.
- **Non-serious AEs** resolved within about 3 weeks: dysuria, haematuria, haemospermia, and urgency or frequency.
- **Crossover arm:** a bladder neck contracture (BNC) with a bladder stone, and urosepsis after cystoscopy.
- **MAUDE (rare):** clot evacuation, transfusion, bladder perforation, epididymitis and infected hydroceles.
- **Rectal safety:** never aim the needle directly posterior (the urethral floor within 1 cm of the verumontanum) when treating a median lobe. **Aim 45° off midline.**
- **Sexual function:**
 - Temporary reduced ejaculate volume or anejaculation occurs early in a few patients and is almost gone by 12 months.
 - **No de novo ED at 5 years** and no significant loss of IIEF or MSHQ-EjD function.

Conclusion

- **WVTT gives durable 5-year efficacy for 30–80 cc, with or without a median lobe.** There is **insufficient evidence** outside that range.
- It can be done under local anaesthesia as an outpatient, with mild-to-moderate AEs and no obvious harm to sexual function.
- Retreatment is comparable or superior to other MISTs, but **probably higher than the reported 4.4%.**

KEY POINTS

- PUL: a MIST for 30–80 cc without a median lobe.
- WVTT: a MIST for 30–80 cc with or without a median lobe.
- Both PUL and WVTT are MISTs for men who want intervention with preservation of erectile and ejaculatory function.

- Warn MIST patients that retreatment within 5 years exceeds that of surgical interventions, and that durability beyond 5 years is uncertain.

COMPARISON — THE TWO GUIDELINE MISTS

	PUL (UroLift)	WVTT (Rezūm)
Principle	Mechanical, non-ablative: permanent implants compress lateral lobes	Thermal ablation by convection: vapour necrosis of transition zone
Volume	30–80 cc (FDA ≤100 cc; AUA insufficient evidence 81–100 cc)	30–80 cc
Median lobe	Excluded	Treatable
Anaesthesia	Local	Oral or conscious sedation, or prostatic block
Catheter	Usually none	Usually a few days (often discretionary)
Early symptoms	Minimal dysuria (mucosa intact)	Irritative symptoms for up to about 3 weeks
Specific hazard	Encrustation if implant <1.5 cm from bladder neck	Rectal injury if needle aimed straight posteriorly near the verumontanum
Key trial	LIFT (vs sham); BPH6 (vs TURP)	Rezūm II (vs sham)
5-year surgical retreatment	13.6%	4.4% (probably underestimated)
Sexual function	Both preserve erectile and antegrade ejaculatory function	

Concept — Convection lets a very small energy dose spread through the interstitium yet stay confined by the capsule. This is why WVTT can ablate a median lobe without the tissue loss and sexual toxicity of resection.

RESECTION-BASED TECHNIQUES

Monopolar Transurethral Resection of the Prostate

Introduction

- **TURP is still the gold-standard surgical treatment.** It has been performed since the 1940s and has a modest learning curve.
- **An electrified wire loop removes transition-zone tissue** from the bladder neck to the verumontanum, down to the depth of the **surgical capsule**.
- **AUA: TURP for small and average prostates (Statement 28).** There is no preference between monopolar and bipolar; choose by expertise (Statement 29).

Mechanism of Action and Technique

- **Monopolar TURP (m-TURP):** current passes from the loop through the tissue and the patient to a grounding pad.
- **It requires non-ionic irrigation** (sterile water, glycine or sorbitol). These fluids are **hypo-osmolar**. Absorption through open sinuses causes acute dilutional hyponatraemia, known as **transurethral resection (TUR) syndrome**. This risk has moved many surgeons to bipolar TURP.

Preoperative considerations.

- **General or spinal anaesthesia.** Spinal was historically preferred so that TUR syndrome could be detected in the awake patient. Signs include confusion, visual loss, nausea and vomiting, electrocardiogram (ECG) changes, coma and death.
- **Positioning:** dorsal lithotomy, padded to protect the **common peroneal nerve**. Put the **buttocks at the table edge**; otherwise the scope collides with the table and the anterior prostate cannot be reached.
- **Baseline abdominal examination** so that later prevesical fluid from a perforation can be recognised.

- **Shaving is not required.** Prepare the lower abdomen, genitalia and perineum.
- **Grounding pad:** on the leg outside the prepared field, on the side opposite any joint replacement.
- **O'Connor rectal shield:** historically allowed sterile rectal access to lift the prostate. It is no longer commercially available.
- **Irrigation:** at body temperature and at the lowest height that gives vision. Raise it only if bleeding obscures the view.

Intraoperative considerations.

- **Before starting:**
 - Check the resectoscope assembly. A camera is standard; continuous flow may be passive or active.
 - Use an organised, systematic plan; the best approach is the one the surgeon knows best.
- **Inspect the bladder first** for tumour or diverticula. An unsuspected tumour is resected first, and TURP is deferred until staging is complete.
- **Note the landmarks and their relation to the adenoma:** the bladder neck, trigone, ureteric orifices, verumontanum and external sphincter. If the orifices are hard to see, give **intravenous indigo carmine or methylene blue**.
- **If the loop does not cut, check in sequence:** scope-to-generator connection → foot pedal → irrigant matches the generator technology → patient properly grounded.
- **A median lobe is resected first**, then the lateral lobes.
- **Current:** mixed (blended) current gives better haemostasis but cuts less well.
- **Approach A — modified Nesbit "encirclement":** start at 11–9 and 1–3 o'clock. Expose, but do not resect, the bladder neck fibres. Carry the resection to the base of the verumontanum, sparing the external sphincter.
- **Fig. 151.6 sequence (anterior start):** 12 o'clock, bladder neck and adjacent adenoma in quadrants → midportion 12 to 9 o'clock → left midfossa 12 to 3 o'clock → midportion 9 to 6 o'clock → apical tissue from beside the verumontanum towards 12 o'clock → residual right side cleared → residual left side cleared, leaving an unobstructed view from verumontanum to bladder.
- **Approach B — floor first (Fig. 151.7):** resect a channel at 5 or 7 o'clock down to the surgical capsule, which sets the depth early → widen laterally → carry up the lateral walls to the anterior aspect along the capsule. The lateral lobes "fall" into the fossa. Finish the 5–7 o'clock floor last to smooth it without undermining the bladder neck.
- **Bladder neck and trigone undermining:** repeated passes across the prostatovesical junction risk it. If the dorsal junction is over-resected early, the scope moves "uphill" and can **detach the trigone** from the prostate base. Set the depth correctly.
- **Chip technique:** long, smooth, "canoe-like" **chips the length of the extended loop**, made with a synchronised rocking motion. Short or thin chips are inefficient and leave an irregular bed that hides bleeders.
- **Anterior and apex last**, for three reasons:
 - The anterior fossa has the least adenoma and perforates easily.
 - The sphincter is tilted, so its anterior part lies more proximally.
 - Emptying the bladder helps to see the anterior area, but keep enough fluid in to avoid perforating the bladder with the loop extended.
- **Premature over-resection or capsular perforation** opens venous sinuses, causing bleeding and fluid absorption.

- **Resect the apex last, in a bloodless field.** A little tissue may be left near the verumontanum: **a repeat resection is preferable to incontinence.**
- **Removing chips:** evacuate all chips with an Ellik or Toomey syringe. Check cellules and diverticula for retained chips, then re-inspect the fossa and control arterial bleeding.
- **Catheter:**
 - Large-bore, using a wire or catheter guide if needed.
 - Tailor the balloon volume to the resected volume so that the balloon does not fall into the fossa.
 - Apply brief traction, released according to haematuria.
- **Continuous bladder irrigation (CBI) is optional.** A three-way catheter does not commit the patient to it, since the inflow port can be plugged. CBI needs skilled nursing, because unrecognised outflow obstruction distends the bladder and worsens bleeding. Intermittent irrigation needs close attention to inflow and outflow.

CONTRAST — ARTERIAL VS VENOUS BLEEDING AT TURP

	Arterial	Venous
Appearance	Bright red, persistent despite irrigation; persists during filling and draining	Darker ; often disappears with a full bladder (fluid pressure tamponades the fossa)
Detection	Visible under flow; bladder-neck (especially anterior) bleeders seen best with the bladder emptied	Check the fossa with the bladder empty and inflow off ; sinuses are hard to find
Control	Must always be controlled: tamponade with the loop, then precise fulguration. "Ricochet" bleeding from the opposite wall or a cryptic site may need deeper resection to expose it; resect to capsule cautiously	Loop tamponade plus fulguration; often cannot be fully controlled → once arterial bleeding is controlled, overfill the catheter balloon (50–60 mL) with traction

Postoperative care.

- **Recovery is usually uneventful:** early diet, and walking on the day of surgery if haematuria is mild.
- **Even an uncomplicated resection absorbs 800–1000 mL of irrigant;** account for this when prescribing fluids.
- **Bladder spasm or rectal discomfort → check the catheter for obstruction.**
- **Catheter out at 24–48 hours** if there is no significant perforation or bleeding. Minor ongoing bleeding allows discharge with close follow-up. Most US surgeons discharge on day 1.
- **Other measures:**
 - **Opioids are almost never needed.**
 - **Stool softeners for 2–4 weeks**, because hard stool can provoke bleeding.
 - **Avoid perineal pressure for 4–6 weeks:** horse riding, cycling, riding lawnmowers.
- **Re-epithelialisation** occurs by migration and proliferation of transitional cells from the margins over a few weeks, so **dysuria is common** during that time. Short-term phenazopyridine may help; warn that it turns body fluids red-orange and stains contact lenses.
- **Warn patients to expect:**
 - **Passage of tissue or eschar with minor delayed bleeding at 1–4 weeks.**
 - Abstinence from sex for at least 2 weeks is commonly advised.
 - Men with long-standing obstruction and storage symptoms may find these persist or worsen. If emptying is confirmed, an anticholinergic or β_3 -agonist can help early. Resolution of detrusor overactivity can take months and is not guaranteed; counsel patience rather than polypharmacy.

Outcomes

- **Most patients rate themselves "better" or "much better"**, depending on baseline severity and follow-up.

- The Agency for Health Care Policy and Research review found that BPH surgery (open prostatectomy, TURP, TUIP) produced about 80% symptom improvement, against 30–40% for placebo and non-surgical therapy.
- **Veterans Administration study (TURP vs watchful waiting, before medical therapy was common):** TURP was clearly more effective at improving symptoms and avoiding failure. Benefit was greatest in men with severe symptoms or high bother.
- **Contemporary data come from m-TURP control arms of RCTs.** Meta-analysis: large gains in Qmax, IPSS, QoL and PVR. **m-TURP had the highest bleeding risk and is the only procedure associated with TUR syndrome.**
- **New technologies rarely beat m-TURP's low surgical retreatment rate.** A national registry showed cumulative secondary TURP rising only slowly over 8 years.
- **Functional results persist at about 10 years,** with slight erosion over time.

Transurethral Resection of the Prostate in the Anticoagulated Patient

- **A dose-escalating, placebo-controlled trial of a heparin-type drug** given at TURP showed dose-dependent blood loss. It was stopped early because bleeding at the highest dose was so severe.
- **The risk of TURP on anticoagulation is significant:** the AUA recommends **HoLEP, ThuLEP or PVP** for patients on anticoagulation (Statement 43).

Complications

- **Intraoperative complications are infrequent** (transfusion, TUR syndrome, bleeding). Overall morbidity approaches about one-fifth when clot retention, recatheterisation, infection, stricture, sexual dysfunction and incontinence are included.

Intraoperative and perioperative complications.

- **TUR syndrome — mechanism:**
 - **Prostatic venous pressure is about 10 mm Hg;** irrigation above this pressure is absorbed through exposed vessels.
 - Hypo-osmolar fluid causes **acute dilutional hyponatraemia.** Features: confusion, nausea, vomiting, visual changes, hypertension, tachypnoea and bradycardia.
- **Risk factors** (Veterans Affairs Cooperative Study): **gland >45 cc and resection time >90 minutes.** The incidence is low and falling in RCT meta-analyses. **Isotonic, iso-osmolar irrigation (bipolar) theoretically eliminates TUR syndrome.**
- **Alternative aetiology — glycine toxicity:** absorbed glycine releases ammonia, giving immediate or delayed encephalopathy.
- **Irrigation height: 60 cm above the patient** (Madsen and Naber) is the minimum for good vision without excess absorption. **An extra 10 cm more than doubles absorption.**
- **Diagnosis:** neurological status plus serum sodium. Check sodium postoperatively after long or large resections, or intraoperatively if concerned.
 - **Na⁺ <120 mEq/L** means significant dilution, with risk of coma, seizures and transtentorial herniation.
 - Transient visual disturbance or blindness indicates central nervous system (CNS) toxicity.
- **Treatment of profound CNS symptoms: judicious hypertonic saline.**
 - **3% saline 100 mL bolus** raises Na⁺ by 2–3 mEq/L. Repeat once or twice at 10-minute intervals if still symptomatic (maximum 300 mL).
 - **Target: raise Na⁺ by 4–6 mEq/L.**
 - A rise of up to 12 mEq/L in the first 24 hours is considered safe for acute post-TURP hyponatraemia.

- If symptoms present **>48 hours after surgery, correct by no more than 8–10 mEq/L per 24 hours.**
- **Overly rapid correction risks central pontine myelinolysis.** Acute post-TURP hyponatraemia is safer to correct than chronic hyponatraemia, because the brain has not yet adapted its organic osmolytes.
- **Ureteric injury is uncommon.** Identify the orifices first, helped by the 70° lens and intravenous methylene blue or indigo carmine. If they are hidden by a high bladder neck or large median lobe, start in the midline and take down the median lobe first; the orifices then appear.
- **Haemorrhage is common.** Achieve full haemostasis to avoid a return to theatre. The transfusion risk is low but must be counselled.
- **Perforation** can occur at the prostatovesical junction, the capsule or the bladder, from resection or from overdistension of thinned capsule.
 - **Glistening periprostatic or perivesical fat is the telltale sign.**
 - If unclear, perform cystography with drainage films.
 - **Prostatic extravasation is almost always extraperitoneal.**
- **Management of perforation:**
 - **Dome perforation → cystogram** to exclude intraperitoneal rupture, which needs **laparotomy and cystorrhaphy.**
 - Extraperitoneal with limited extravasation → prolonged catheter drainage and observation.
 - Extensive extravasation → percutaneous or open drainage.
- **Intraoperative erection** limits scope movement.
 - It often subsides spontaneously; otherwise give **intracorporeal phenylephrine** and alert anaesthesia to possible cardiovascular effects.
 - If injections fail, **perineal urethrostomy** is a last resort.

Postoperative complications.

- **Bladder neck contracture:**
 - Occurs in a small minority. It is caused by **over-resection plus excessive fulguration of the bladder neck**; an undermined trigonal flap may also heal as a membrane.
 - **Typical story:** excellent early flow that slowly declines, more markedly than with regrowth. Average onset is about 6 months (range 3 weeks to 10 years). Serial flow rates decline.
- **BNC management — step-up:**
 1. **Office cystoscopy** to confirm.
 2. **Gentle dilation** with sounds or a balloon.
 3. **Endoscopic incision** with a Collins knife, optical urethrotome or laser, over an open-ended catheter or wire, until the scar springs open. If the opening is obliterated, intravenous methylene blue with suprapubic pressure shows a blue jet, usually anterior. Incise **aggressively, down to perivesical fat**, because after TURP the sphincter is distant; contrast the anastomotic stricture after radical prostatectomy, where the sphincter is close.
 4. **Recurrent BNC → repeat transurethral incision (TUIBNC) plus perilesional mitomycin C** (10 mL of 0.4 mg/mL).
 5. **Intractable BNC → open or robotic Y-V plasty.** Endoscopic suturing revision is a newer option.
- **Urethral stricture:**
 - Occurs in a small minority, from resectoscope trauma or postoperative infection.
 - **Choose an appropriate sheath size.** Many men do not calibrate to ≥ 28 Fr at the meatus and fossa navicularis.

- Calibrate or gently dilate the meatus, inspect on insertion and **lubricate generously**. The authors instil 30 mL of gel and dilate with Van Buren sounds to at least 2 Fr above scope size.
- **Incontinence:**
 - **The internal sphincter is the main male continence mechanism**, and TURP resects it or renders it incompetent. **Continence then depends entirely on the external sphincter**; injuring it causes total stress urinary incontinence (SUI).
 - **The verumontanum is the key landmark**, both as the distal stop point and for future cystoscopy. Resection ending proximal or adjacent to it rarely damages the sphincter.
 - The sphincter complex is poorly demarcated, thinnest anteriorly, and tilted so that its anterior part lies more proximally.
 - **SUI after TURP is rare and usually transient**, rarely needing intervention.
- **Postoperative haemorrhage:**
 - **Significant early bleeding usually reflects incomplete haemostasis.**
 - **Delayed bleeding at 1–4 weeks with eschar is common.** It becomes less likely with time, and mild bleeding settles with rest and fluids.
- **Postoperative retention** occurs in a small minority, a proportion unchanged over time. Prolonged catheterisation reflects bleeding or surgeon preference.
- **Retreatment:**
 - Causes: incomplete resection (apex or anterior), **poor selection (gland >80 cc)**, technical error, stricture, BNC, regrowth or residual BPH.
 - Retreatment for BPH is rare. Revision risk rises with **gland size and preoperative retention**.
- **Ejaculatory dysfunction:** the bladder neck is resected, so **retrograde ejaculation affects about two-thirds**. Erectile, orgasmic, desire and satisfaction domains are not worsened by either monopolar or bipolar TURP. Discomfort on ejaculation is not increased.
- **Erectile dysfunction:** theoretically possible through thermal spread to the neurovascular bundles. However, temperatures there stay near body temperature during TURP. **There is no conclusive evidence that TURP causes ED.**

Conclusion

- **m-TURP is durable and effective** for IPSS, QoL and objective parameters, and it is the **benchmark for every new BPH therapy**. Its morbidity drove the development of less invasive alternatives.

Concept — *Every characteristic hazard of monopolar TURP follows from **current returning through the patient**. That forces hypo-osmolar irrigation, which causes TUR syndrome, which rewards speed over haemostasis. Continence, by contrast, depends on one landmark: stopping at the verumontanum.*

BIPOLAR TRANSURETHRAL RESECTION OF THE PROSTATE

Introduction

- **Bipolar TURP (b-TURP) carries the active and return poles on the same electrode**, so current stays at the resection site and does not pass through the patient to a pad.
- This allows **isotonic, iso-osmolar saline irrigation, eliminating TUR syndrome**. Large resections and meticulous haemostasis are possible without time pressure.
- **True bipolar:** a dual-loop design in which energy arcs between adjacent loops while cutting.

- **Transurethral Resection in Saline (TURis) was not truly bipolar:** the outer sheath acted as the return electrode, exposing the whole urethra and penis to return energy. Strictures and urethrocutaneous fistulae resulted, and it fell out of favour.
- **Plasma system (Olympus; common in the US):**
 - Energy vaporises the saline around the loop; further energy converts the gas to plasma. **Excited sodium ions give the orange glow.**
 - It resects at lower voltage and temperature while sealing vessels, giving better haemostasis.
 - The loop is **platinum-iridium, not tungsten**, to withstand thermal and electrical stress.
 - The coagulation setting does not create plasma; it heats tissue to seal vessels.
 - Energy spread is narrower and charring is less, possibly giving fewer storage symptoms.

Technique

- **Essentially identical to m-TURP.** It is often quicker, because cutting and sealing are simultaneous.
- **Plasma ignition:** older generators struggled to ignite the plasma; newer generators have largely solved this. **Drag on the loop at first tissue contact means the plasma glow has not ignited.**

Outcomes

Comparative Studies

- **Repeated meta-analyses of RCTs: m-TURP and b-TURP have equivalent efficacy** for IPSS, QoL and PVR, in both the short term and the long term (up to 5 years).
- **Qmax may favour b-TURP slightly.** The difference is statistically but not clinically significant, and is skewed by a few heavily weighted trials.

Complications

- **The key safety gain comes from isotonic irrigation.** The "cut and seal" energy profile and the removal of time pressure improve haemostasis.

Perioperative

- **Meta-analyses: b-TURP has fewer overall complications, with lower transfusion, clot retention, TUR syndrome (none), immediate reoperation and haemoglobin drop.**
- **No difference in UTI or AUR.**
- **Number needed to treat with b-TURP:** about 50 to prevent one TUR syndrome, and about 20 to prevent one clot retention.

Postoperative

- **Urethral stricture, BNC, incontinence, erectile and ejaculatory dysfunction, and BPH retreatment are broadly similar** between modalities. Some series show slightly less stricture and BNC with bipolar; one earlier meta-analysis showed non-significantly more stricture with bipolar.
- **Catheter duration and length of stay (LOS) are shorter with b-TURP.**

Conclusion

- **b-TURP is equivalent in efficacy but safer**, and is likely to replace m-TURP as the gold standard. Better haemostasis and isotonic irrigation allow longer, safer resections.

CONTRAST — MONOPOLAR VS BIPOLAR TURP

	Monopolar (m-TURP)	Bipolar (b-TURP)
Current path	Loop → patient → grounding pad	Active and return poles on the loop; stays local

	Monopolar (m-TURP)	Bipolar (b-TURP)
Irrigant	Non-ionic, hypo-osmolar (water, glycine, sorbitol)	Normal saline (isotonic, iso-osmolar)
TUR syndrome	Yes — only procedure associated with it	Theoretically eliminated
Loop	Tungsten	Platinum-iridium (Plasma)
Bleeding / transfusion / clot retention	Higher	Lower
Catheter time, LOS	Longer	Shorter
IPSS, QoL, PVR	Equivalent (Qmax marginally favours bipolar)	
Stricture, BNC, retreatment, sexual function	Broadly similar	

TRANSURETHRAL INCISION OF THE PROSTATE

Introduction and Concept

- **Incising the prostate or bladder neck dates to Guthrie (1834).** TUIP disrupts the prostatic capsule; capsular contraction has been proposed as a cause of LUTS.
- **TUIP only for small glands (<30 cc; Statement 31).** The ideal patient is a **young man with a small prostate concerned about ejaculation or fertility.**
- **Retrograde ejaculation risk is lower than with TURP,** especially with a unilateral incision. If ejaculation is paramount, avoid the bladder neck and a complete capsular incision.
- Because MISTs also preserve ejaculation with minimal morbidity, **TUIP may be replaced.** It remains reasonable where MISTs are unavailable.

Technique

- **Selection is critical:** large glands or severe symptoms respond poorly and will need more treatment.
- **A median lobe is a relative contraindication,** since most studies excluded it. The EAU recommends TUIP for <30 cc **without a median lobe.**
- **Instruments:** a hot (Collins) knife, a thin resectoscope loop, or an end-firing holmium or thulium laser.
- **Sequence (Fig. 151.8):** avoid the ureteric orifices → unilateral or bilateral incision at **5 and 7 o'clock** → start distal to the ureteric orifice → through the bladder neck → end proximal to the verumontanum → deepen until the **surgical capsule** is seen (some go on to periprostatic fat) → haemostasis with cautery.
- **Bleeding should be minimal.** The catheter is removed and the patient discharged on day 0–1. Convalescence is rapid, with few irritative symptoms.

Outcomes

Comparative Studies — TUIP versus TURP

- **The trials are of poor to moderate methodological quality** and are restricted to small glands, which limits meta-analysis.
- **TUIP gives shorter operating time, less blood loss and shorter LOS.** Catheter duration was longer after TUIP in one trial.
- **Symptom scores improve similarly,** with a small advantage for TURP. **Qmax improves more after TURP.**
- **Reoperation is higher after TUIP.** Residual obstructing lobes or interlobar adhesions are common at follow-up cystoscopy.
- **Holmium laser TUIP in glands <30 cc gives durable improvement** to about 5 years.

Complications

Intraoperative and Perioperative

- **Transfusion is rare** and much less frequent than after TURP; some TURP comparators had implausibly high transfusion rates.
- **Capsular perforation**, intended or not, is managed with prolonged catheterisation.
- **Temporary retention** occurs in a minority.

Postoperative

- **Retrograde ejaculation is less likely than after TURP** (reported 0–37%, probably at the lower end). A single incision may carry less risk, but other reports found two incisions did not increase it.
- **No difference from TURP** in retention, infection, stricture, incontinence or ED.
- **Reoperation is more likely after TUIP.**
- **BNC was not seen after TUIP**, whereas it occurred after TURP. Urethral stricture was uncommon.

Conclusion

- **TUIP gives adequate results in well-selected men:** a short, simple operation with minimal stay and less retrograde ejaculation. **Discourage it for moderate or large glands or a median lobe.**

KEY POINTS

- m-TURP remains an important treatment and the benchmark for new BPH technologies.
- b-TURP's key advance is resection in isotonic irrigation: no TUR syndrome and better bleeding outcomes.
- AUA 2023: m-TURP or b-TURP for <30 cc and 30–80 cc.
- TUIP is reasonable in carefully selected men (<30 cc, wishing to preserve antegrade ejaculation); a median lobe is a relative contraindication.

Concept — *TUIP relieves obstruction by releasing the capsule and bladder neck rather than removing adenoma. That buys ejaculatory preservation and speed at the cost of durability, and confines it to small glands without a median lobe.*

ABLATION-BASED TECHNIQUES

Robotic Waterjet Therapy

Introduction and Concept

- **RWT (Aquablation)** uses a **non-thermal, high-velocity saline waterjet** to ablate tissue.
 - The surgeon maps the treatment area on real-time transrectal ultrasound (TRUS).
 - A robotically guided transurethral jet then ablates it in minutes.
 - The bladder neck, external sphincter and periprostatic nerves are spared from thermal or anatomical injury.
- **Short learning curve, high capital cost.** Proponents claim minimal effect on continence, erection and antegrade ejaculation.
- **AUA: RWT for 30–80 cc (Statement 39)**, based on an RCT. **A median lobe can be treated.**

Technique

- **Aquabeam Robotic System:** a console and pump, a robotic transurethral handpiece and a TRUS probe. **Requires general or spinal anaesthesia.**

- **Sequence:** dorsal lithotomy → biplanar TRUS in the rectum → 24 Fr handpiece with integrated cystoscope, fixed to a bed-rail arm, placed 1–2 cm past the bladder neck → retract the cystoscope to confirm position → sagittal TRUS view capturing prostate, bladder and handpiece → register the handpiece (centred on the transverse view, rotated between 3 and 9 o'clock) → plan depth and angle on the console, **excluding the bladder neck and external sphincter** → foot pedal fires the jet in proximal-to-distal sweeps.
- **Waterjet velocity controls ablation depth**, to a maximum of 2.4 cm. Treatment is monitored by TRUS and cystoscopy.
- **The sphincter is protected** because ablation occurs only in front of the cystoscope.
- **Limits:**
 - **RWT ablates only posterior tissue between 4 and 8 o'clock**, so it is unsuitable for substantial anterior tissue.
 - **No histological specimen.**
- **Haemostasis:** originally by Foley traction, which was associated with more bleeding. **Focal bladder-neck cautery is now recommended:**
 - With a resectoscope, bluntly clear the "fluffy" ablated remnant tissue with the loop, then cauterise the exposed vessels. Aggressive resection is unnecessary because the tissue is already ablated.
 - This markedly reduced transfusion.
 - The EAU advises informing patients of the bleeding risk.

Outcomes

Comparative studies.

- **WATER:** an international, **double-blind** multicentre RCT of RWT vs TURP, 2:1.
 - Inclusion: age 45–80, 30–80 cc on TRUS, IPSS ≥ 12 , Qmax < 15 mL/s, PVR < 300 mL.
 - Designed for IPSS non-inferiority, with a margin of 4.7 points.
- **At 6 months RWT was non-inferior.** Its mean IPSS improvement was 1.8 points greater, and **more so in the 50–80 cc subgroup**. QoL, Qmax and PVR did not differ.
- **Total operating time was similar, but resection time was far shorter** with RWT (about 4 vs 27 min).
- **Durability:** equal IPSS gains at 1 year. At 3 and 5 years improvements were durable and equivalent, **with larger IPSS gains for RWT in 50–80 cc glands**.

Complications

Perioperative.

- **WATER:** a greater haemoglobin drop with RWT; one transfusion after RWT and none after TURP. Hospital stay and catheter duration were equal.
- **WATER II (large glands, non-comparative):** a high 30-day transfusion rate and prolonged catheter traction. **Focal bladder-neck cautery has since reduced bleeding.**
- **Rectal injury is rare but catastrophic;** one required a diverting colostomy. Place the TRUS probe carefully and consider **routine rectoscopy after RWT**.

Postoperative.

- **WATER at 3 months: fewer complications with RWT (26% vs 42%),** driven by fewer Clavien grade 1 events.
- **Retrograde ejaculation was lower with RWT (10% vs 36%).** This was attributed to the absence of thermal energy and to planning that avoids the verumontanum.
- **Trap — the true rate is probably higher.** Men with baseline anejaculation, possibly from α -blockers, were excluded from the count, and the TURP comparator rate is unexpectedly low.

- **Sexual function:** IIEF-15 erectile, desire, orgasmic and intercourse domains were equal; overall satisfaction was better with RWT. MSHQ-EjD was better at 6 months and remained better, including bother, at 3 years.
- **Medication discontinuation:** most men on α -blockers or 5 α -reductase inhibitors (5ARIs) stopped them in both arms; none started new BPH drugs by 6 months.
- **Retreatment:**
 - **WATER:** one-year complications did not differ; the only BNC was after TURP; surgical retreatment at 3 years was 4.3% RWT vs 1.5% TURP (not significant). Five-year retreatment data lacked detail.
 - **WATER II (large prostates):** 5-year medical retreatment 6% and surgical 3%, with none in years 4–5.

Conclusions

- **AUA: RWT for 30–80 cc.** It needs general or spinal anaesthesia, so it is **not a MIST**.
- **Non-inferior to TURP**, and possibly superior for 50–80 cc.
- **Large-gland (>80 cc) data are non-comparative** and so are not in the AUA Guidelines.
- Focal bladder-neck cautery has reduced the early bleeding concern.

Concept — *RWT is heat-free, image-planned removal of posterior adenoma. Absent heat explains the better ejaculatory profile, image planning explains the short learning curve, and the absence of cautery explains the early bleeding problem.*

TRANSURETHRAL VAPORISATION OF THE PROSTATE

Introduction and Concept

- **TUVP (Kaplan and Te, 1995) uses TURP equipment with a larger-surface electrode.** Higher energy spread over a broader contact area **vaporises rather than resects**, with simultaneous coagulation.
 - This gives better haemostasis and a clearer field.
 - No new capital outlay is needed.
- **VaporTrod (Olympus; Fig. 151.9):** a grooved roller electrode, the most-studied monopolar option.
- **TUVP moved to bipolar energy in the late 1990s**, allowing isotonic irrigation. **AUA: bipolar TUVP for small and average glands (Statement 32).**
- **PlasmaButton (Olympus; Fig. 151.10):** a hemispherical electrode with a large surface area for current accumulation; an updated version vaporises a larger area. Plasma is generated as in b-TURP, but **tissue contact is almost always needed to create the orange "corona"**.
- **All tissue is vaporised, so no pathological specimen is obtained.**

Outcomes

- **AUA and EAU both recommend only bipolar TUVP (b-TUVP).**
- **Monopolar TUVP (m-TUVP) fell out of favour:** early results matched TURP, but durability was questioned. **Coagulation spread up to 10 mm** caused irritative voiding symptoms and stress incontinence.

Comparative Studies — Bipolar Studies

- **b-TUVP vs m-TURP RCTs:** b-TUVP had shorter operating and catheter times and no transfusions. At 12 months IPSS improved more with b-TUVP, but the TURP arm had performed unusually poorly; Qmax was equal.
- **Durability reverses the picture:** at 2–3 years and at 6 years, **IPSS and Qmax favoured TURP**, and **retreatment (surgical and medical) was higher after b-TUVP**. Single-centre data with heavy attrition bias, but they question durability.
- **Meta-analyses:** no difference in IPSS, QoL or PVR; shorter catheter time with TUVP; a trend towards better Qmax with TURP.

Complications

Intraoperative and Perioperative

- **Simultaneous vaporisation and coagulation reduce bleeding.** Transfusion is lower, though this does not reach significance because it is rare in both arms.
- Capsular perforation and conversion to TURP are essentially absent. **Clavien 3–4 complications are fewer than after TURP.**

Postoperative

- **Retrograde ejaculation is common** (the great majority with the VaporTrove), with no change in erectile function. ED and retrograde ejaculation rates equal TURP.
- **Among the highest rates of recatheterisation for AUR**, and higher surgical retreatment and dysuria than TURP or HoLEP.

Conclusion

- **b-TUVP is a viable alternative to TURP** with excellent haemostasis and vision. Enthusiasm was tempered by **durability (retreatment) and dysuria**. AUA: b-TUVP for <30 cc and 30–80 cc, not larger.

KEY POINTS

- RWT is recommended by the AUA for 30–80 cc.
- RWT may preserve sexual function better than TURP, but published results may overestimate preservation of antegrade ejaculation.
- Bipolar TUVP is recommended by the AUA for <30 cc and 30–80 cc.

Concept — *Vaporisation trades haemostasis for a coagulated, specimen-free bed. The coagulum explains the dysuria and retention, and the absence of a clean resection plane explains the weaker durability. The same trade recurs with PVP.*

LASER PHYSICS

- **Laser stands for light amplification by stimulated emission of radiation.** Laser energy is converted to local heat that destroys tissue.
- **Advantages:** ambulatory (<23-hour observation) or same-day surgery, and minimally invasive versions of invasive operations, such as HoLEP replacing open simple prostatectomy (OSP).
- **Coagulation lasers were abandoned for vaporisation.** The laser coagulation code (Current Procedural Terminology, CPT 52647) is essentially obsolete compared with laser vaporisation (CPT 52648).

CONTRAST — COAGULATION VS VAPORISATION

	Coagulation	Vaporisation
Tissue temperature	About 60 °C	≥100 °C (boiling)
Effect	Protein denaturation → coagulative necrosis	Intracellular water boils → immediate destruction
Tissue death	Delayed, with sloughing	Immediate

- **Laser parameters:**
 - **Wavelength** (nm): the distance between waves.
 - **Pulse energy** (J): the work per pulse.
 - **Pulse frequency** (Hz): pulses per second.
 - **Power** (W): energy per unit time.

- **The tissue effect depends on wavelength and on the absorption coefficient of the target.** Wavelength sets penetration depth (Fig. 151.11).
- **The water absorption peak is 1940 nm.** The thulium fibre laser (TFL) emits at 1940 nm, so its energy is absorbed about four times more readily in water than holmium energy, giving faster vaporisation.
- **Potassium-titanyl-phosphate (KTP) laser, 532 nm:** absorbed selectively by haemoglobin and not by water, giving immediate, efficient, haemostatic vaporisation.

FIG. 151.11 OPTICAL PENETRATION DEPTH BY LASER, WITH CHROMOPHORE (REBUILT)

Laser	Wavelength	Chromophore	Optical penetration
Green light — KTP / lithium triborate (LBO)	532 nm (visible)	Haemoglobin	0.8 mm
Diode	—	—	0.5–5 mm (deepest)
Holmium:YAG (Ho:YAG)	≈2120 nm (2.1 μm)	Water	0.4 mm
Thulium:YAG / TFL	2013 nm / 1940 nm	Water (TFL at the absorption peak)	0.2 mm (shallowest)
Neodymium:YAG (Nd:YAG)	1064 nm	—	About 1 cm (in text; abandoned)

- **What the table tests:** the water-absorbed infrared lasers (Ho, Tm) penetrate least. Green light penetrates further but is haemoglobin-targeted, and Nd:YAG penetrates deepest of all, which is why it caused sloughing.
- **Laser safety:**
 - **Wavelength-specific protective eyewear for everyone in theatre, including the patient,** and available at the door.
 - Occupational Safety and Health Administration (OSHA)-approved warning signage outside the theatre.
 - Windows and portals blocked.
 - Laser inspected before each case, and all operators trained.
 - An institutional laser safety officer available for advice.

Concept — Chromophore determines behaviour. *Water-absorbed lasers cut and enucleate with shallow, controllable heat; the haemoglobin-absorbed green laser vaporises bloodlessly; and deep penetration means coagulum and sloughing.*

PHOTOSELECTIVE VAPORISATION OF THE PROSTATE

Introduction and Concept

- **PVP uses KTP or LBO lasers, derived from Nd:YAG.**
- **Nd:YAG was abandoned:** its 1 cm penetration caused massive sloughing, intermittent voiding and retention, and it raised safety concerns.
- **Passing the 1064 nm Nd:YAG beam through a KTP or LBO crystal doubles its frequency and halves the wavelength to 532 nm.** This is absorbed by haemoglobin, an intravascular target, with no energy lost to water.
- **Higher energy density** gives preferential vaporisation with a **thin (0.2 mm) coagulation rim**, and so good haemostasis.
- **Platforms** ("Greenlight" is a trade name):
 - **KTP 80 W and 100 W.**
 - **LBO 120 W High Performance System (HPS) and 180 W (XPS).**

- **The 180 W platform** has the same power density and the same 1–2 mm vaporisation and coagulation depth as 120 W, with a **larger beam area** (0.28 → 0.44 mm²). The MoXy fibre adds:
 - A metal tip against devitrification.
 - Built-in fluid cooling and automatic overheating protection.
 - Pulsed coagulation.
- **Higher power vaporises more efficiently** in animal, ex vivo and clinical studies. Vaporisation and coagulation are simultaneous, closing sinuses and preventing fluid absorption.
- **AUA: PVP (120 or 180 W) for <30 cc and 30–80 cc (Statement 33)**, and an option for **higher bleeding risk (Statement 43)**.

Technique

Pre- and Intraoperative

- **Standard workup.** PVP is acceptable when anticoagulation cannot be stopped.
- **Equipment:** a 23–24 Fr rigid cystoscope. **A 600 µm side-firing fibre emits at 70° to the long axis;** the 180 W system uses a 750 µm fibre.
- **Method:** sweep the fibre over the parenchyma, vaporising successive superficial layers until **capsular fibres are seen**.
- **Bladder-neck sequence (first step): identify the ureteric orifices** → lower power (about 80 W) in this area if preferred → one midline incision or incisions at 5 and 7 o'clock to open the bladder neck → vaporise the posterior bladder neck until level with the trigone → turn the fibre medially or laterally for vision.
- **Safety at the bladder neck:**
 - Aim laterally, never towards either ureteric orifice.
 - **Avoid aggressive coagulation at the bladder neck.**
 - Some spare an anterior strip of urothelium to prevent BNC.
- **Posterior tissue is harder than at TURP**, because the fibre cannot treat tissue directly in line with it; this matters most with a high bladder neck. Vaporise posterior tissue in **horizontal layers, never aiming posteriorly towards the rectum**. Move the fibre more than the scope.
- **Working distance:**
 - Too close gives contact vaporisation and fibre damage.
 - Too far gives inefficiency and **coagulation instead of vaporisation**.
 - Use lower power in tight spaces, such as kissing lateral lobes. Prevent tissue build-up on the fibre.
- **Sweep technique:**
 - Continuous, even sweeps avoid irregular craters: too fast gives poor vaporisation, too slow creates a crater.
 - The most efficient sweep angle is 15–30°, with least coagulation at 30°.
 - Novice rule: **sweep 30° in 1–2 seconds**.
- **Large bubbles signal efficient vaporisation.** Their absence means coagulation or charring, which wastes energy and increases dysuria and sloughing.
- **Bleeding hot spots:** apex, median lobe, bladder neck and posterolateral prostate, where vessels enter.
 - Moderate bleeding → vaporise the adjacent tissue to control feeding vessels.
 - Pulsatile bleeding → coagulation mode (about 30 W) around and on the vessel. Pinching the vessel with the scope or fibre tip before coagulating its edges helps.

- Refractory bleeding → **Bugbee electrode** through the bridge (requires switching to a **non-ionic irrigant**), then a larger sheath and resecting loop.
- **Finish:** re-inspect the bladder for errant laser injury, place a catheter and confirm clear irrigation.

Postoperative

- **Catheter out on the day of surgery or the next day.** Some discharge the same day with self-removal at home next day. **Tissue passage with minimal bleeding at 7–10 days;** manage minor bleeding with fluids and follow-up.
- **Dysuria is proportional to the volume of coagulated tissue**, and worse with inefficient technique.
 - Risk factors: large median lobe, prior prostatitis, dense or fibrous tissue, previous TUNA or TUMT.
 - **Sterile pyuria is expected;** treat with non-steroidal anti-inflammatory drugs (NSAIDs) or a short steroid taper.
 - Always exclude a tissue flap or retained fibre fragment.
- **The residual coagulum can cause subacute prostatitis or UTI.** If infection is suspected, give culture-directed antibiotics and consider a test of cure.

Outcomes

- **Only 120 and 180 W data are relevant** to the guideline. PVP is effective and safe.
- **Learning curve:** 30–50 cases for familiarity, **and more than 100 for efficient vaporisation.**

Comparative Studies

- **GOLIATH (landmark):** a non-inferiority RCT of **180 W PVP vs TURP** at 29 European centres.
 - Men aged 40–80, IPSS ≥ 12 , Qmax ≤ 15 mL/s, **prostate ≤ 100 cc**, American Society of Anesthesiologists (ASA) score ≤ 3 , PVR ≤ 400 mL. Monopolar or bipolar TURP was allowed.
 - About 270 randomised.
- **Results:**
 - **PVP non-inferior for IPSS, Qmax and complications at 6 months, 1 year and 2 years.** There was no 5-year follow-up.
 - **PVP:** shorter catheter time, time to stable health and LOS, and fewer early (≤ 30 -day) interventions.
 - **TURP:** shorter operating time.
 - Self-reported incontinence was higher after PVP at 6 months but equal by 1 year.
 - **Retrograde ejaculation was similar (about two-thirds in each arm)**, and reoperation at 2 years was equal.
- **Meta-analysis of 120 W PVP vs m-TURP:** no difference in IPSS, Qmax or PVR. TURP was faster. PVP had less transfusion, shorter catheter time and LOS, and no difference in BNC or stricture, but **higher surgical retreatment.**
- **>80 cc — not recommended.** An RCT of 80–150 cc glands found high overall retreatment after PVP and TURP but minimal after HoLEP. Large-gland PVP series show rising surgical retreatment.
- **Energy density:** reoperation is lower with 180 W than with lower-power platforms. **>5 kJ/cc is associated with greater PSA decline**, a marker of more complete vaporisation.

Photoselective Vaporisation in the Anticoagulated Patient

- **AUA: PVP for patients at higher bleeding risk (Statement 43).**
- Men on aspirin, warfarin or clopidogrel improve as much as others, with **no meaningful increase in transfusion, reoperation, recatheterisation or perforation.** Catheter duration is slightly longer.

Complications

Intraoperative and Perioperative

- **No TUR syndrome has ever been reported**, because saline is used. **Transfusion is very rare**, even on anticoagulants, and markedly lower than after m-TURP.
- **Capsular perforation is rare** (0.2–1%). Maintain an even sweep and deepen circumferentially and systematically; the capsule is harder to see early in the learning curve.
- **Ureteric orifice injury**: identify the orifices first, aim away from them, and do not advance the fibre too far into the bladder.

Postoperative

- **Dysuria and storage symptoms are common and self-limiting** within 3 months. PVP has the **highest dysuria rate** among transurethral techniques. Anti-inflammatories, culture-directed antibiotics or a short steroid course help.
- **Infection**: UTI and epididymitis are more frequent because the necrotic coagulum harbours bacteria. **Consider longer antibiotic courses.**
- **Stricture**: the 24 Fr sheath (vs 26–28 Fr for TURP) may theoretically lower stricture risk. Meta-analysis shows no significant difference in stricture or BNC.
- **Erectile function is minimally affected overall**. Men with normal or mildly impaired baseline function may decline, though none needed ED treatment.
- **Retrograde ejaculation is common**, similar to TURP.

Conclusion

- **Vaporisation replaced coagulation**. Higher-power platforms improved efficiency and retreatment, and 532 nm haemoglobin selectivity allows treatment of anticoagulated patients. AUA: **<30 cc, 30–80 cc and high bleeding risk; not >80 cc**, where retreatment is unacceptably high.

KEY POINTS

- Laser vaporisation has supplanted laser coagulation.
- PVP exploits selective absorption of 532 nm by haemoglobin, giving efficient vaporisation and haemostasis.
- AUA: PVP (120 or 180 W) for small and average prostates (<30 cc and 30–80 cc).
- AUA: PVP for patients at higher risk of bleeding.

Concept — Frequency-doubling converts a deep, sloughing Nd:YAG beam into a **haemoglobin-seeking 532 nm beam**. Haemostasis and anticoagulant safety follow from this, while the coagulum it leaves behind explains PVP's dysuria, infection and size ceiling.

ENUCLEATION-BASED TECHNIQUES: ANATOMIC ENDOSCOPIC ENUCLEATION OF THE PROSTATE

- **"Enucleation is enucleation is enucleation is enucleation."** Whatever the energy source, a true anatomic endoscopic enucleation of the prostate (AEEP) gives excellent, durable outcomes.

Holmium Laser Enucleation of the Prostate

Introduction and Concept

- **Ho:YAG (≈2120 nm) is highly absorbed by water.**
 - It is pulsed, with **0.4 mm penetration**, and transmits along flexible quartz fibres.

- It vaporises with a superficial coagulation zone. Heat dissipates over 2–3 mm and coagulates small and medium vessels.
- **Evolution:**
 - **Holmium laser ablation of the prostate (HoLAP, 1994)**, which was slow for large glands.
 - **Holmium laser resection of the prostate (HoLRP)**: strips pushed into the bladder and removed with a grasper.
 - **HoLEP (1998)**, made practical by the **tissue morcellator**.
- **HoLEP is the endoscopic equivalent of open simple prostatectomy** and the most technically advanced laser prostate operation. Variants include top-down, bottom-up and early apical release; the whole adenoma is enucleated en bloc or in 2–3 lobes, which are pushed into the **bladder** and morcellated.
- **HoLEP is the most rigorously studied laser technique**, with excellent, durable results and minimal early and late morbidity.
- **Its steep learning curve limits uptake**. Estimates start at 20 supervised cases, but efficiency plateaus after about 50 cases or later; one self-taught surgeon had not plateaued by 125 cases.

Technique

Preoperative.

- **HoLEP is size-independent**: <30, 30–80 and >80 cc, with or without a median lobe (Statement 38).
- **Antibiotics**: many cover gram-positive organisms (ampicillin) and gram-negative organisms with good prostatic penetration (gentamicin).
- Performed in hospital or a surgery centre under **regional or general anaesthesia**.

Intraoperative.

- **Equipment:**
 - **End-firing 550 µm fibre**, through a continuous-flow laser resectoscope (24–28 Fr) with a 30° lens.
 - The fibre is stabilised in a 7 Fr stabilising catheter or a 5–6 Fr open-ended catheter in the bridge.
 - **Normal saline irrigation**.
- **60–150 W Ho:YAG**. A dual pedal switches instantly between enucleation and coagulation, though it has not been shown to speed enucleation. Pulse modulation (Moses; Virtual Basket or Vapor Tunnel) may shorten operating time.
- **Typical settings: 2.0 J × 40–50 Hz for enucleation; 1.0–2.0 J × 20 Hz for coagulation**.
- **Three-lobe bottom-up sequence with early apical release:**
 1. **Median lobe present**: bladder-neck incisions at 5 and 7 o'clock down to the trigone and to capsule depth. The capsule's **reflective, smooth surface is the depth reference** for the rest of the case. Carry the incisions distally to just proximal to the verumontanum, join them horizontally, and enucleate the median lobe upwards into the bladder without undermining the bladder neck.
 2. **No median lobe**: a single 6 o'clock incision instead.
 3. **Perimontanal inverted-U incision**; the scope pushes laterally to develop the lateral lobe–capsule plane at the apex.
 4. Carry the plane anteriorly: clockwise to 11–12 o'clock on the right, then counterclockwise to 12–1 o'clock on the left.
 5. **Incise the "apical mucosal strip"** to avoid traction on the urinary sphincter (the modified early apical release).

6. Join the anterior incisions and develop the anterior plane proximally. Vertical bladder-neck fibres (the "**Baleen**" sign) herald bladder entry at 12 o'clock; incise the bladder neck and drop its lateral wings.

7. Divide the anterior commissure from distal to proximal, making two lobes (three with a median lobe). Release the lateral attachments and enucleate the lobes into the bladder.

8. **Haemostasis** with a defocused beam on coagulation settings. Good haemostasis gives good vision for morcellation.

- **Morcellation:**

- Swap the inner sheath for a long nephroscope with a 5 mm working channel.
- **Two inflow lines, fully open, keep the bladder distended.**
- Morcellators have suction and cutting modes: Piranha (rotating blades); Versacut (forward-back oscillating); Multicut Solo (rotating and oscillating). Fragments are collected in a trap for histology.

- **Morcellator bladder injury is rare** with good vision and a full bladder.

- If mucosa is engaged: **stop, stop suction, disengage** (disassembling the blades if necessary). Most injuries are superficial and need catheter drainage only.
- **Intraperitoneal perforation → laparotomy and cystorrhaphy.**

- **Access problems:** rarely, a very large adenoma or morbid obesity needs **perineal urethrostomy** to reach the bladder.

- **Catheter: 22 Fr three-way, with about 60 mL in the balloon** so it stays at the bladder neck and does not slip into the fossa. A Mandrin guide or placement over a wire avoids posterior capsular perforation.

Postoperative.

- **Traditionally overnight, with catheter out on day 1.** With modern lasers and technique, **same-day discharge and same-day catheter removal succeed in most patients.**

- **A large capsular perforation → keep the catheter for a few days**, then remove it in clinic without further concern.

Outcomes

HoLEP versus TURP.

- **RCT of glands <100 cc:**

- HoLEP gave greater improvement in AUASS and PVR, with equal Qmax.
- It had **less haemoglobin drop, shorter catheter time and shorter LOS**, but a **longer operating time**.

- **Meta-analyses:**

- HoLEP is superior for IPSS and Qmax (and PVR in one), with equal QoL and no difference in complication rates.
- It has less sodium and haemoglobin fall and shorter catheter time and LOS; TURP is faster.
- The advantage in IPSS and Qmax is durable at 3–8 years.
- **HoLEP is the only endoscopic procedure with efficacy superior to TURP.**

HoLEP in the anticoagulated patient.

- **Evidence is from small retrospective cohorts** of men on anticoagulant or antiplatelet therapy (AC/AP).

- **Findings:**

- Enucleation and morcellation are no slower.
- **LOS and CBI are slightly longer.** Transfusion is similar in most series, but higher on antiplatelets in one.
- Continuing AC/AP carries a non-significant trend to more transfusion than holding it.

- More 90-day complications, though not emergency visits or major (Clavien ≥ 3) events.
- **Warfarin fared worse than direct oral anticoagulants:** fewer successful voiding trials, and more complications and emergency visits.
- **Counsel** about longer stay, longer CBI and possible transfusion; functional outcomes are equal. **AUA: HoLEP for higher bleeding risk (Statement 43).**

Complications

Intraoperative and perioperative.

- **Morcellation** speeds HoLEP compared with HoLRP but introduces **morcellator bladder injury**, which is rare in experienced hands. Prevent it with a distended bladder and haemostasis before morcellating.
- **Retained enucleated adenoma** does not degrade or pass. It causes irritative symptoms, bleeding and infection and needs repeat morcellation. It occurs after morcellator malfunction, or when poor vision leads the surgeon to think morcellation is complete.
- **Ureteric orifice injury** is uncommon and concentrated in glands >80 cc. Identify the orifices first and aim away when making the 5 and 7 o'clock incisions.
- **Capsular perforation:**
 - **Too shallow** leaves residual adenoma, causing bleeding from uncontrolled feeding vessels and later recurrence.
 - **Too deep** causes perforation. Reported rates vary with definition, since some count "threatened" perforation with fat visible. Even full-thickness perforation heals with a few days of catheter drainage.
- **Haemorrhage is very low:** transfusion is rare and conversion to TURP essentially never occurs. The haemoglobin drop is smaller than after TURP and increases with gland size.

Postoperative.

- **Storage symptoms and urgency are expected**, from laser energy on the capsule. Transient dysuria is uncommon, and most symptoms are mild.
- **Transient incontinence is common early but rarely persists.** It is similar after OSP, which suggests the sheer volume of adenoma removed, not transurethral instrumentation, is responsible. Persistent SUI is rare.
- **Early apical release** to reduce sphincter traction:
 - Incise the prostatic urethral mucosa at 11–1 o'clock → incise the paracollicular mucosa → join to the anterior incision, releasing the apex from the sphincter.
 - Early SUI falls steadily over 6 months.
 - It is usually paired with en bloc enucleation. With two- or three-lobe techniques, know where the sphincter is when splitting the commissure and releasing the apical strips, or **permanent incontinence** may result.
- **BNC:**
 - Uncommon (0–3.2%) and **more frequent in smaller glands.**
 - Size thresholds reported: **<40 cc (prophylactic bladder-neck incision advised), <54 cc and <55 cc.**
 - A 28 Fr scope did not significantly increase BNC compared with 26 Fr. Rates equal OSP.
- **Urethral stricture:**
 - Uncommon, associated with **longer operating and catheter times**, and most often at the **meatus.**
 - No difference between 26 and 28 Fr scopes; scopes ≤ 24 Fr are unstudied.
 - Direct vision internal urethrotomy (DVIU) rates are similar to OSP.

- **Retreatment for BPH is rare** because the whole transition zone is removed (Fig. 151.12). It occurred occasionally over 18 years, in **glands <55 cc** and early in the learning curve.
- **Retrograde ejaculation is common (about three-quarters)**, because the whole transition zone is removed.
 - The orgasmic domain falls; desire and satisfaction do not.
 - **"Median-lobe-only" HoLEP preserves antegrade ejaculation in about 90%**; its durability is not yet known.
- **No negative effect on erectile function**; scores are unchanged or slightly improved.

FIG. 151.12 ZONAL ANATOMY — WHAT IS REMOVED

State	Transition zone	Consequence
Pre-operative BPH	Enlarged transition zone and median lobe; central and peripheral zones compressed; anterior fibromuscular stroma anterior	Obstruction
After HoLEP (enucleation)	Completely removed to the capsule	Negligible regrowth; retreatment rare
After TURP	Partially removed ; peripheral rim of adenoma remains	Regrowth possible

Conclusion

- **HoLEP is size-independent (Statement 38)**, with excellent, durable results comparable or superior to TURP. It has less bleeding, shorter catheter time and LOS, less transfusion, and similar BNC and stricture rates. It is recommended for **higher bleeding risk (Statement 43)**.

THULIUM AND THULIUM FIBRE LASERS

Concept and Introduction — Comparison of Thulium and Thulium Fibre Lasers

- **Thulium:YAG (Tm:YAG)**: a solid-state, diode-pumped laser at **2013 nm**. It was originally continuous-wave only, which with its **0.2 mm penetration** caused carbonisation. It can now run continuous or pulsed.
- **TFL: 1940 nm**, continuous or pulsed. Water absorbs this about **four times more than holmium energy**, giving efficient ablation. Continuous-wave thulium is said to give a "cleaner" incision than holmium.
- **Thulium techniques**:
 - Thulium vaporessection (ThuVAP).
 - Thulium vaporisation (ThuVAP).
 - **Thulium vapoenucleation (ThuVEP)**, which uses more laser energy to "vapo-incise".
 - **ThuLEP**, which uses mainly **blunt dissection**, with the laser only for mucosal incision, lobe division, releasing attachments and haemostasis.
- **ThuLEP matches HoLEP (Statement 38: size-independent)**. Proponents claim a **shorter learning curve** because the continuous wave cuts precisely and shows the plane.
 - Claims of 8–16 cases are not supported, since efficiency had not plateaued.
 - Simulator-trained novices needed about 30 cases.

Technique

- **ThuLEP (described in 2010)**: incise and enucleate the transition zone, then morcellate exactly as for HoLEP.
 - Equipment: 26 Fr laser scope, Piranha morcellator, 550 µm fibre.
 - Settings: 70–90 W for enucleation, 30 W for coagulation. For thulium fibre enucleation (ThuFLEP): 1 J × 60 Hz short pulse to enucleate; 1 J × 30 Hz long pulse to coagulate.
- **Steps mirror the three-lobe HoLEP technique**. Catheter at least overnight, usually out on day 1 once haematuria resolves; discharge after a successful trial without catheter (TWOC).

Outcomes

ThuLEP versus TURP

- **RCT (no size limit):** equal improvement in IPSS, QoL, Qmax and PVR at 3 months. ThuLEP gave **less resected weight, longer operating time and less haemoglobin drop**, with equal catheter time and LOS.
- **Meta-analysis (thulium resection vs TURP):** equal functional outcomes; thulium was **slower but had less bleeding, shorter catheter time and LOS, and fewer complications**.

ThuLEP versus HoLEP

- **RCT of ThuVEP vs HoLEP (median about 80 cc):** equal functional improvement. ThuVEP had **faster enucleation** (equal total time), fewer overall complications and less postoperative retention.
- **Confounders:** holmium without pulse modulation, set at about 40 W against thulium at 90 W.
- **Meta-analysis:** slightly less haemoglobin drop and less transient incontinence with ThuLEP, but the **certainty is low**. Postoperative retention in some series is higher than typical HoLEP benchmarks.

ThuLEP in the Anticoagulated Patient

- **Small retrospective series only.** With AC/AP continued (often aspirin), catheter time and stay were longer, transfusion and clot retention were uncommon, and no operative intervention for bleeding was needed.
- ThuVARP with continued vs held AC/AP gave equal operative time, catheter time, stay, haemoglobin drop and function.
- **ThuLEP is acceptable on continued AC/AP (Statement 43).**

Complications

Intraoperative and Perioperative

- **Safe in experienced hands.** There was no incomplete morcellation or ureteric injury; **morcellator bladder injury is uncommon**, similar to ThuVEP. Transfusion is uncommon.

Postoperative

- **Early morbidity:** symptomatic UTI and early retreatment for residual BPH occur in a minority. **Mild storage symptoms are frequent but settle within a month.** Transient urge incontinence resolves by 1 year.
- **Late complications:** BNC and urethral stricture are uncommon at 4–5 years. **Retreatment for recurrence is low at 5 years.**
- **Sexual function:** no effect on erectile function (IIEF, IIEF-5); retrograde ejaculation in most (about four-fifths) of those with baseline antegrade ejaculation.

Conclusion

- **Thulium 2013 nm (continuous) and TFL 1940 nm (quasi-continuous).** Continuous-wave thulium may carbonise tissue, which does not impair recovery beyond the learning curve. **ThuFLEP** is newer but promising.
- **ThuLEP is size-independent (Statement 38) and suitable for higher bleeding risk (Statement 43).**
- **HoLEP and ThuLEP have the strongest evidence** among AEEP energy sources. Long-term follow-up is needed to show whether others match HoLEP.

KEY POINTS

- HoLEP and ThuLEP are options for patients at increased bleeding risk (Statement 43).
- HoLEP and ThuLEP are size-independent, with excellent results (Statement 38).
- HoLEP has long-term outcomes comparable to those historically seen with open prostatectomy.
- Thulium enucleation was popularised later; longer-term (>15-year) outcomes are still needed.

- Alternative energy sources (KTP/LBO, bipolar or plasmakinetic, diode) can perform AEEP satisfactorily, including large glands, but need comparative studies.

COMPARISON — HOLMIUM VS THULIUM ENUCLEATION

	HoLEP	ThuLEP / ThuVEP
Wavelength / mode	≈2120 nm, pulsed	Tm:YAG 2013 nm continuous (or pulsed); TFL 1940 nm
Penetration	0.4 mm	0.2 mm
Water absorption	High	About 4× holmium
Dissection style	Laser-driven enucleation	ThuLEP mainly blunt; ThuVEP laser "vapo-incision"
Learning curve	Steep (≥20; plateau ≥50 or later)	Claimed shorter (about 30)
Evidence base	Largest ; follow-up to 18 years	Shorter follow-up
Guideline status	Both size-independent (Statement 38); both for higher bleeding risk (Statement 43)	
Efficacy / sexual function	Equivalent; retrograde ejaculation common; erectile function preserved	

Concept — *Enucleation removes the entire transition zone along the anatomical capsular plane. This single fact gives size independence, near-zero regrowth, ejaculatory loss, and a BNC risk concentrated in small glands. The energy source matters far less than the plane.*

TRUE MINIMALLY INVASIVE SURGICAL THERAPY CONCEPT

- **MISTs still need rigid cystoscopy** (PUL, WVTT), which is uncomfortable.
- **True minimally invasive surgical therapies (TMISTs):**
 - Off-the-shelf devices deployed by **flexible cystoscopy without general anaesthesia or sedation**.
 - Preserve sexual function and improve symptoms.
 - Offer an alternative to lifelong medication.
- **They are deliberately not durable:** the target durability is **12–18 months**.

TEMPORARY IMPLANTED PROSTATIC DEVICES

Concept and Technique

- **TIPD (iTind; Olympus; originally Medi-Tate, Israel)** is FDA-approved. It gives immediate relief without a permanent implant.
- **Design:**
 - **Three elongated nitinol struts at 12, 5 and 7 o'clock.**
 - An **anchoring leaflet at 6 o'clock** against migration.
 - A polyester retrieval suture.
 - Expands to a maximum 33 mm diameter and 50 mm length.
- **Deployment:** by flexible or rigid cystoscope, under local anaesthesia or oral or intravenous sedation.
- **Mechanism:** **left in place for 5–7 days**, exerting radial force that remodels the bladder neck and prostatic urethra by **ischaemic pressure necrosis**.
- **Removal:** advance an 18 Fr silicone catheter over the suture and pull it taut, so the device collapses into the catheter.
- **AUA 2023:** **TIPD is an option for 25–75 cc without an obstructing median lobe (Statement 41).**

Outcomes

- **Pivotal RCT, TIPD vs sham (catheter), 2:1.**
 - Inclusion: age ≥ 50 , IPSS ≥ 10 , Qmax ≤ 12 mL/s, 25–75 cc, no obstructing median lobe.
 - Primary end point: **proportion with ≥ 3 -point IPSS improvement at 3 months.**
- **Results:**
 - **Met:** a higher proportion of responders with TIPD than sham, including at a ≥ 7 -point threshold. Almost all responders remained responders at 12 months.
 - **But the absolute IPSS difference from sham at 3 months was not significant**, and QoL did not differ. Qmax improved.
 - Tolerability was good; AEs were mild and lasted 2–4 weeks.
 - A small minority needed alternative surgery or restarted medication by 12 months.
 - **No de novo erectile or ejaculatory dysfunction.**
- **Single-arm 2-year study:** durable IPSS, QoL, Qmax and PVR with no sexual dysfunction. **A median lobe (treated despite exclusion) predicted failure.**

Conclusion

- **TIPD met its primary end point but not a significant absolute IPSS difference.** There are no long-term data and no comparative RCT against TURP or other MISTs. **The AUA statement rests on expert opinion only.**

DRUG-COATED BALLOON DILATION

Concept and Technique

- **Optilume BPH Catheter** has two balloons:
 - A **predilation balloon** delivers mechanical force, dilating the channel and causing an **anterior commissurotomy**.
 - A **paclitaxel-coated balloon** delivers an antiproliferative drug to prevent regrowth and commissural scarring. It is inflated for at least 5 minutes.
- The balloon design anchors force in the **prostatic fossa, not the bladder neck**. A catheter is left for at least 48 hours.

Outcomes

- **EVEREST-1:** prospective, single-arm, open-label, multicentre (Latin America).
 - Inclusion: age ≥ 50 , IPSS ≥ 13 , Qmax 5–15 mL/s, PVR ≤ 250 mL, **20–80 cc, prostatic urethral length 35–55 mm.**
 - Excluded: prior transurethral surgery, **median lobe >1 cm**, stricture and neurogenic bladder.
 - Primary end point: $\geq 40\%$ IPSS improvement at 3 months.
- **Results:**
 - The primary goal was exceeded, with IPSS and QoL **improving by about two-thirds to 12 months** and a meaningful Qmax gain.
 - Complications occurred in a sizeable minority: mainly haematuria, incontinence and UTI. **Ejaculatory dysfunction occurred in a small minority.**
- **Pending:** 4-year EVEREST-1 data and the sham-controlled, double-blind PINNACLE RCT. EVEREST-1 was done under general or spinal anaesthesia; the aim is office-based use as a TMIST.

Conclusion

- **TMISTs could improve QoL for men who do not want daily medication.** TIPD fully preserves sexual function and has one sham RCT. Long-term safety and efficacy data are needed for all TMISTs.

COMPARISON — TEMPORARY AND MINIMALLY INVASIVE DEVICES

	PUL	WVTT	TIPD (iTind)	Drug-coated balloon (Optilume)
Category	MIST	MIST	TMIST	TMIST (aspirational)
Mechanism	Permanent compressive implants	Convective thermal ablation	Temporary nitinol struts; ischaemic pressure necrosis over 5–7 days	Dilation / anterior commissurotomy + paclitaxel
Volume	30–80 cc	30–80 cc	25–75 cc	20–80 cc (trial)
Median lobe	Excluded	Treatable	Excluded (predicts failure)	Excluded if >1 cm
Cystoscope	Rigid	Rigid	Flexible or rigid	—
Evidence	RCT vs sham and vs TURP; 5 years	RCT vs sham; 5 years	One sham RCT; AUA statement is expert opinion	Single-arm, 1 year; not in AUA list
Sexual function	Preserved	Preserved	Preserved	Ejaculatory dysfunction in a small minority

PROSTATE ARTERY EMBOLISATION

Introduction and Concept

- **First reported in 2000.** Planning may use MRI or CT pelvic angiography.
- **Sequence:** femoral arterial access (usually **unilateral right femoral**) under local anaesthesia → pelvic angiography of the iliac tree and prostatic arteries → urethral catheter with the balloon filled with 30% contrast and 70% saline as a landmark → 1.9–2.4 Fr microcatheters into the prostatic arteries → embolic agent (e.g. alcohol or microspheres) infused until **stasis**.
- **Bilateral PAE gives better results** than unilateral.
- **Contraindications:**
 - Acontractile or underactive bladder, or neurogenic bladder.
 - Urethral stricture or BNC.
 - Large bladder diverticula or stones.
 - Contrast allergy and chronic kidney disease.
- **Technically demanding:**
 - Tortuosity, atherosclerosis and aberrant anatomy can prevent completion.
 - The prostatic artery arises from five different trunks, **most often the internal pudendal artery**. There are often two prostatic arteries on one side, and anastomoses are common.
 - Procedure and fluoroscopy times are long, giving substantial **radiation exposure**.
- **Guidelines:**
 - **EAU:** for moderate-to-severe LUTS in men who **accept less-than-optimal outcomes compared with TURP**. Urologists do the workup and follow-up with trained interventional radiologists.
 - **AUA 2021:** not outside clinical trials.
 - **AUA 2023:** may be offered, by trained clinicians, after discussion of risks and benefits (Statement 40).

Data

Comparative Studies

- **RCT of PAE vs m-TURP at 2 years:** TURP superior for IPSS, QoL, Qmax and PVR, with greater falls in PSA and volume.
- **UK Register of Prostate Embolisation (UK-ROPE):** a propensity-matched comparison with TURP. **PAE was not non-inferior (it was inferior) for IPSS and QoL**, but LOS was shorter (median 0 vs 2 days).
- **Meta-analyses:** TURP gives greater Qmax and volume reduction. IPSS, QoL, IIEF-5 and PVR are similar, and PAE has **fewer AEs and shorter LOS**. A Cochrane review found little or no short-term difference in symptoms or QoL.

Complications

Intraoperative

- **Access site:** pseudoaneurysm, bleeding and **arterial dissection**; dissection may delay embolisation but has no other sequelae.
- **Non-target embolisation:** bladder or rectal ischaemia, usually managed conservatively. **Penile discoloration and small ulcers** can occur through anastomoses.

Postoperative

- **Common minor events:** UTI, transient haematuria, urethral burning, rectal bleeding and haemospermia.
- **One RCT vs b-TURP:** more complications with PAE, including **AUR needing catheterisation** in about a quarter, post-embolisation syndrome and treatment failure. Another RCT vs m-TURP found fewer complications with PAE, with similar severity and none of Clavien grade ≥ 4 .
- **Durability is poor.** By 10 years a majority need surgical reintervention (repeat PAE or TURP), and medical retreatment is common. About a fifth of trial patients had TURP within 2 years.
- **Sexual function:** IIEF is stable or improved, and **ejaculatory dysfunction is less than after TURP**. The UK-ROPE retrograde ejaculation figure could not separate pre-existing from new cases.

Conclusion

- **PAE is low-morbidity, outpatient, and effective but inferior to TURP.** There is no size-based recommendation. The AUA notes a role in **refractory haematuria secondary to BPH**. It should be performed only by trained clinicians (Statement 40).

Concept — PAE treats the gland **without entering the urinary tract**. That explains its low morbidity and preserved ejaculation, and also its dependence on vascular anatomy, its weaker efficacy and its high long-term reintervention.

CONCLUSION

- **Many MISTs are available and more are coming**, differing in invasiveness, morbidity, durability and degree of tissue removal. New technologies bring new complications.
- **Scrutinise new studies:** designs lacking rigour or sham control may overstate results.
- **The gold-standard operation for a given man is the one that meets his needs and expectations safely.** It is chosen by shared decision-making around his most bothersome symptoms and goals, patient factors, and the surgeon's familiarity with and ability to perform the procedure.

COMPARISON — NAMED TRIALS AGAINST TURP OR SHAM

Trial	Intervention vs comparator	Population	Verdict
LIFT	PUL vs sham	30–80 cc	Superior to sham; retreatment 13.6% at 5 years

Trial	Intervention vs comparator	Population	Verdict
BPH6	PUL vs TURP	≤60 cc	Composite: non-inferior and superior; TURP better on IPSS, Qmax and PVR
Rezūm II	WVTT vs sham	30–80 cc, median lobe allowed	Superior; durable at 5 years; no de novo ED
WATER	RWT vs TURP (double-blind)	30–80 cc	Non-inferior; better in 50–80 cc; less retrograde ejaculation
WATER II	RWT, single-arm	Large glands	Low 5-year retreatment; high early transfusion
GOLIATH	180 W PVP vs TURP	≤100 cc	Non-inferior to 2 years; shorter catheter time and stay
Pivotal TIPD RCT	TIPD vs sham	25–75 cc	Responder end point met; absolute IPSS difference not significant
EVEREST-1	Drug-coated balloon, single-arm	20–80 cc	Primary goal met; 1-year data only
UK-ROPE	PAE vs TURP (matched registry)	—	Inferior for IPSS and QoL; shorter stay

AUA BPH GUIDELINE STATEMENTS CITED IN THIS CHAPTER

Statement	Content
1 / 3	History, examination, IPSS and urinalysis at baseline / repeat IPSS after treatment
5 / 6 / 7 / 8	Consider size and shape imaging / PVR required / consider uroflowmetry / pressure-flow if uncertain
9	Counsel about treatment failure and retreatment
26	Indications for surgery
28 / 29	TURP for small and average glands / monopolar or bipolar by expertise
31 / 32 / 33	TUIP <30 cc / bipolar TUVP small and average / PVP (120 or 180 W) small and average
36 / 37	WVTT 30–80 cc / WVTT to preserve erectile and ejaculatory function
38	HoLEP or ThuLEP, size-independent
39 / 40 / 41	RWT 30–80 cc / PAE by trained clinicians / TIPD 25–75 cc, no obstructing median lobe
43	Higher bleeding risk → HoLEP, ThuLEP or PVP